Evaluating an AI-Based Skin Cancer Diagnostic Test

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A new medical test uses artificial intelligence (AI) to help doctors find skin cancer early. One example is DermaSensor, an AI-enabled device recently authorized by the U.S. Food and Drug Administration (FDA) for use in primary care to detect skin cancers such as melanoma, basal cell carcinoma, and squamous cell carcinoma (Venkatesh, 2024). This test helps non-specialist doctors check suspicious moles by analyzing light and image data through machine learning algorithms.

This AI diagnostic is designed to look at skin lesions and decide if they are likely cancerous. The AI uses deep learning, a type of neural network that can learn patterns from thousands of skin images. Deep learning models, especially Convolutional Neural Networks (CNNs), are strong for recognizing patterns in medical images (Salinas et al., 2024).

Performance of diagnostic tests is measured by sensitivity and specificity. Sensitivity is how well the test identifies people who truly have the disease. Specificity is how well the test identifies people who do not have the disease. Recent research shows that AI skin cancer tools can reach high levels of accuracy similar to trained clinicians, with many models achieving sensitivity and specificity that compare well with dermatologists (Salinas et al., 2024).

However, AI tools are not perfect and may still give false positives or false negatives, meaning the test might sometimes falsely indicate cancer or miss a real case. Because of this, clinicians usually use AI as a support tool, not a replacement for doctors.

Patients who might need this test include those with suspicious moles, people at high risk for skin cancer, or individuals in remote areas without easy access to specialists. In my view,

this AI test is good because it can help find cancer early and support clinical decisions, but it should always be paired with doctor expertise.

References

Salinas, M. P., et al. (2024). *A systematic review and meta-analysis of artificial intelligence for skin cancer classification*. *npj Digital Medicine.* https://doi.org/10.1038/s41746-024-01103-x

Venkatesh, K. P. (2024). *Learnings from the first AI-enabled skin cancer device for primary care use*. *npj Digital Medicine.* https://doi.org/10.1038/s41746-024-01161-1